

PP1 Panel Preparation Guide

Project R26-IT-117 — AI-Based Architectural Planning Model

Component 01 — Architectural Planning API

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Component	Component 01 — Architectural Planning API
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1. Project Overview

Project R26-IT-117 is an AI-driven construction planner for Sri Lankan residential construction. Component 01 is a FastAPI service that takes a Sri Lankan cadastral (survey) plan as input and automatically generates 3 floor plan alternatives as SVG/PDF output, following NBC Sri Lanka building regulations.

System Input / Output

Input	Output
Cadastral plan (PDF/JPG/PNG/TIFF)	3 Floor plan alternatives (SVG + PDF)
User requirements (rooms, floors, style)	Building Schema JSON for Component 02
Setback inputs (road type, margins)	Validated site schema with GPS coordinates

2. The 4-Stage Pipeline

Stage	Name	Endpoint	Type
1	Cadastral Plan Extraction	POST /api/v1/extract	Synchronous
2	Buildable Zone Calculation	POST /api/v1/buildable-zone	Synchronous
3	AI Floor Plan Generation	POST /api/v1/generate-floor-plan	Async (Celery)
4	Render SVG/PDF	POST /api/v1/render	Synchronous

Stage 1 — Cadastral Plan Extraction

Uploads a cadastral plan image/PDF. Runs EasyOCR to read all text, OpenCV to detect the plot boundary polygon, spaCy+regex NER to extract structured fields (plan number, district, area, scale, surveyor, coordinates), and pyproj to convert SLD99 survey coordinates to WGS84 GPS. Output is a validated Site Schema JSON.

Stage 2 — Buildable Zone Calculation

Applies NBC Sri Lanka setbacks (front/rear/side margins) and BCR (Building Coverage Ratio) based on district and coastal flag. Uses Shapely to compute the legal buildable polygon. Returns max footprint area, recommended floors, and orientation.

Stage 3 — AI Floor Plan Generation (Async)

RAG retrieves 5 similar Sri Lankan house designs from Supabase pgvector using SentenceTransformer embeddings. PromptBuilder constructs a structured prompt with site constraints, user requirements, and RAG context. Gemini 2.5 Flash is called 3 times at temperatures 0.4/0.7/1.0 to produce conservative/balanced/creative alternatives. Each plan is validated for geometric violations and scored on 4 dimensions. Celery+Redis handles async execution.

Stage 4 — Render SVG/PDF

SVGRenderer draws the floor plan with room rectangles, dimension lines (metres), window indicators, north arrow, and title block. PDFRenderer generates a 6-page professional report. SchemaSerialiser validates the Building Schema JSON and fire-and-forgets a POST to Component 02 /estimate.

3. Models & Technologies Used

Model / Library	Stage	Purpose
EasyOCR (CRAFT + CRNN)	Stage 1	Reads text from cadastral plan images
PyTorch CNN (ResNet-based)	Stage 1	Classifies cadastral vs non-cadastral plans
OpenCV (Canny + contours)	Stage 1	Detects plot boundary polygon
spaCy en_core_web_sm + regex	Stage 1	NER — extracts district, area, coordinates
pyproj (EPSG:5234 → 4326)	Stage 1	Converts SLD99 coordinates to WGS84 GPS
Shapely	Stage 2	Geometric polygon operations and area calculation
NBC Rule Engine (custom)	Stage 2	Sri Lanka National Building Code setbacks
all-MiniLM-L6-v2 (SentenceTransformer)	Stage 3	384-dim embeddings for RAG vector search
Supabase pgvector	Stage 3	Cloud vector database for RAG similarity search
Gemini 2.5 Flash (Google)	Stage 3	LLM — generates 3 floor plan alternatives
Shapely (validator)	Stage 3	Checks room overlap violations
svgwrite	Stage 4	Draws SVG floor plan diagram
ReportLab	Stage 4	Generates professional PDF report
jsonschema	Stage 4	Validates Building Schema JSON contract
FastAPI	API	High-performance REST API framework
Celery + Redis	Stage 3	Async task queue for 60-90s Gemini generation
Docker	Infra	Containerisation for deployment

4. Dataset & Data Collection

Component	Dataset	Status
EasyOCR	Pre-trained multilingual model (ICDAR, SynthText datasets)	Ready — no fine-tuning needed
CNN Classifier	Collecting 100+ Sri Lankan cadastral plans from licensed surveyors	Surveyors collecting from father (surveyor)
NER Parser	Rule-based + spaCy base. Sri Lankan patterns: 26 districts, 96 geotags, 1000+ units	Working on D90 geotags, pending units
RAG Knowledge Base	10 Sri Lankan coastal residential house descriptions	Expanded to 50 pages collecting from civil eng
Gemini 2.5 Flash	Google pre-trained LLM — prompted with NBC Sri Lanka regulatory hints and RAG context	Ready for AI hinting RAG context

CNN Classifier — 5 Classes

Class	Description
surveyor_plan	Standard licensed surveyor cadastral plan
municipal_plan	Municipal council subdivision plan
uda_plan	Urban Development Authority plan
low_quality	Scan too poor for reliable extraction
non_cadastral	Not a cadastral plan — reject

5. Accuracy & Performance Metrics

Component	Metric	Value / Status
EasyOCR	Text recognition accuracy on printed cadastral plans	85.05% on standard printed text
CNN Classifier	Classification accuracy (5 classes)	Pending — fine-tuning on 100+ collected plans. Expected ~90% accuracy
NER Parser	Field extraction accuracy on standard cadastral formats	95% on standard SL cadastral plan formats
Coordinate Conversion (pyproj)	SLD99 → WGS84 accuracy	Mathematically exact (EPSG:5234 standard)
RAG Retriever	Similarity search — passages retrieved per query	5 passages per query, cosine similarity > 0.7
Floor Plan Scorer	Quality scoring dimensions	4 dimensions: space utilisation, natural light, adjacency
Stage 3 Celery Task	End-to-end generation time	60-170 seconds (3 Gemini API calls)
Test Suite	Unit + integration tests passing	31/31 tests passing

6. Expected Panel Questions & Answers

Q: What AI/ML processes are in your system?

7 AI/ML processes: (1) Computer Vision — OpenCV edge detection for boundary polygon. (2) Deep Learning OCR — EasyOCR CRAFT+CRNN neural network. (3) CNN Classification — ResNet-based cadastral plan classifier. (4) NLP/NER — Named Entity Recognition for structured data extraction. (5) RAG — vector similarity search + LLM generation. (6) LLM Prompting — Gemini 2.5 Flash with structured prompt engineering. (7) Geometric Validation — Shapely polygon intersection detection.

Q: You said you use pre-trained models — is that allowed?

Yes. We use pre-trained models as base and extend them for Sri Lankan domain. EasyOCR is extended with Sri Lankan cadastral text patterns. spaCy base model is extended with Sri Lankan NER rules (26 districts, SLD99 coordinates, perch units). CNN classifier is being fine-tuned on 100+ real Sri Lankan cadastral plans. Gemini is prompted with Sri Lankan NBC regulations and RAG context from local data.

Q: What dataset did you use?

We have 4 data sources: (1) Pre-trained EasyOCR — no dataset needed. (2) CNN — collecting 100+ Sri Lankan cadastral plans from licensed surveyors. (3) NER — rule-based on Sri Lankan specific patterns (26 districts, SLD99 coords). (4) RAG — 10 Sri Lankan coastal house descriptions in Supabase, expanding to 50+.

Q: What is your accuracy?

EasyOCR: 85-95% on printed cadastral text. NER: ~95% on standard formats. Coordinate conversion: mathematically exact. CNN: fine-tuning in progress, expected 90%+ after training on 100+ collected plans. 31/31 unit tests passing.

Q: Why did you use RAG?

RAG gives Gemini real Sri Lankan house design context. Without RAG, Gemini generates generic Western-style floor plans. With RAG, it generates designs suited for Sri Lankan climate — cross-ventilation, shade, coastal pile foundations, verandahs. RAG retrieves the 5 most similar designs from Supabase pgvector using cosine similarity.

Q: Why 3 temperature values for Gemini?

Temperature controls creativity in LLM output. 0.4 (Conservative) — low randomness, strictly follows NBC constraints. 0.7 (Balanced) — moderate creativity, balances rules and innovation. 1.0 (Creative) — high randomness, more innovative designs. This gives the user 3 real alternatives to choose from.

Q: Why Celery and Redis?

Gemini takes 60-170 seconds to generate 3 floor plans. A synchronous HTTP request would time out. Celery runs the generation as a background task, the user gets a task_id immediately and polls /floor-plan-status/{task_id} for the result.

Q: Why Supabase instead of a local vector database?

Supabase pgvector is cloud-hosted PostgreSQL with vector extension — no local ChromaDB to manage or deploy. Works from anywhere, persistent storage, and integrates with standard SQL for structured queries alongside vector search.

Q: What is SLD99 and why do you convert it?

SLD99 (EPSG:5234) is Sri Lanka's official local coordinate system used by all licensed surveyors. GPS uses WGS84 (EPSG:4326). We convert so the plot location can be displayed on Google Maps and used by other system components. pyproj handles the mathematical transformation.

Q: What NBC regulations do you follow?

NBC = National Building Code of Sri Lanka. Front setback: 3m (national road), 1.5m (lane). Rear setback: 1.5m minimum. Side setback: 1m minimum. BCR (Building Coverage Ratio): Colombo 0.60, Coastal districts 0.40 (stricter), Other districts 0.50. Coastal sites also require pile foundations.

Q: How does your component connect to other components?

Input: Receives cadastral plan upload from frontend (Component 04). Output: Sends Building Schema JSON to Component 02 (cost estimation) via HTTP POST to /estimate endpoint. If Component 02 is unavailable, Stage 4 logs a warning and continues — the render is never blocked by C02 availability.

Q: What are the limitations of your system?

1. CNN fine-tuning in progress — collecting real Sri Lankan cadastral plans. 2. Floor plan geometry may have overlaps — LLM does not guarantee non-overlapping rooms. Post-processing geometric solver planned for next sprint. 3. Currently tested primarily on Ampara district data — expanding to all 25 districts. 4. Generation takes 60-170 seconds due to 3 Gemini API calls.

7. NBC Sri Lanka — Quick Reference

Road Type	Front Setback	Notes
National Road / Main Road	3.0 m	Primary arterial roads
Provincial Road	2.0 m	Secondary roads
Local Road	1.5 m	Residential streets
Lane / Private Road	1.0 m	Narrow access lanes

Building Coverage Ratio (BCR) by District

District Type	BCR	Max Coverage
Colombo (urban)	0.60	60% of land area
Other Districts (standard)	0.50	50% of land area
Coastal Districts (Ampara, Galle, etc.)	0.40	40% of land area — stricter

Foundation Type Rules

Condition	Foundation Type
Coastal site (is_coastal = True)	Pile foundation
Large plot (area > 500 sqm)	Raft foundation
Standard plot	Strip foundation

8. System Architecture & API Endpoints

Method	Endpoint	Description
POST	/api/v1/extract	Stage 1 — Upload cadastral plan, get Site Schema
POST	/api/v1/buildable-zone	Stage 2 — Compute buildable polygon from site schema
POST	/api/v1/generate-floor-plan	Stage 3 — Dispatch async floor plan generation
GET	/api/v1/floor-plan-status/{task_id}	Stage 3 — Poll for generation result
POST	/api/v1/render	Stage 4 — Render SVG + PDF + Building Schema
GET	/api/v1/download/{filename}	Download rendered SVG or PDF file
GET	/api/v1/health	Health check — returns status ok

Infrastructure Components

Service	Technology	Purpose
API Server	FastAPI + Uvicorn (port 8001)	REST API — handles all HTTP requests
Task Queue	Celery 5.4 + Redis	Async background tasks for Stage 3
Vector DB	Supabase pgvector	RAG knowledge base — cosine similarity search
LLM	Gemini 2.5 Flash (Google AI Studio)	Floor plan generation — 3 alternatives
Container	Docker	Deployment containerisation

9. What to Demo at PP1

Demo Flow (Step by Step)

Step	Action	Expected Output
1	Show GET /api/v1/health	{"status": "ok", "component": "C01"}
2	POST /api/v1/extract with a cadastral plan	Building Schema JSON with plan_id, district, area_sqm, boundary_polygon, GPS coordinates
3	POST /api/v1/buildable-zone with site boundary	Buildable polygon, max footprint sqm, setback margins
4	POST /api/v1/generate-floor-plan	task_id returned immediately
5	GET /api/v1/floor-plan-status/{task_id}	3 or plan alternatives — conservative, balanced, creative
6	POST /api/v1/render with chosen plan	SVG floor plan + PDF report + Building Schema JSON
7	Open the SVG file	Visual floor plan with room dimensions, north arrow, title block
8	Open the PDF file	6-page professional report with quality scores, room schedule

Key Points to Highlight

- Full 4-stage pipeline working end-to-end
- AI generates 3 alternatives at different creativity levels (temperatures)
- RAG uses real Sri Lankan house design knowledge
- NBC Sri Lanka regulations automatically applied
- SLD99 → WGS84 GPS coordinate conversion for Sri Lankan survey plans
- Async processing with Celery so API never times out
- 31/31 unit tests passing
- Building Schema sent to Component 02 for cost estimation
- CNN classifier and NER fine-tuning in progress with real data collection